The Periodic Table

Exam code: 0620 & 0971

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The Periodic table

- There are over 100 chemical elements which have been isolated and identified
  - Each element has one proton **more** than the element preceding it
  - This is done so that elements end up in columns with other elements which have similar properties
  - Elements are arranged on the periodic table in order of **increasing atomic number**
  - The table is arranged in vertical columns called **groups** and in rows called **periods**

The period number

- Periods are the horizontal rows that show the number of shells of electrons an atom has and are numbered from 1 - 7
- Elements in Period 2 have two electron shells, elements in Period 3 have three electron shells

The group number

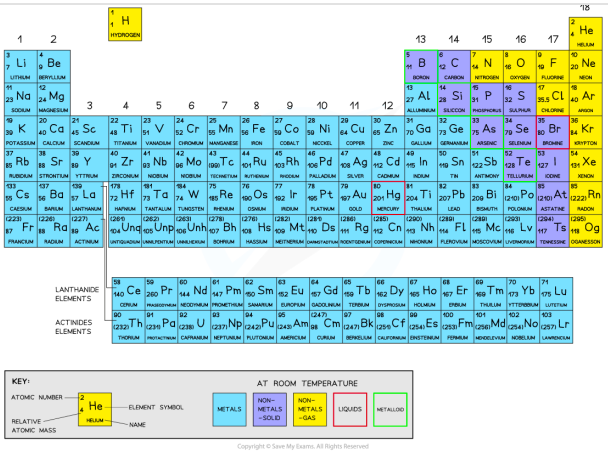
- Groups are the vertical columns that show how many outer electrons each atom has
- They are numbered from 1–7 with a final group called Group 0 (instead of Group 8)
- Group 4 elements have atoms with 4 electrons in the outermost shell, Group 6 elements have atoms with 6 electrons in the outermost shell and so on
- The group number can help determine the charge that metal and non-metal ions form
- For metals, the group number corresponds to the number of electrons it will lose to achieve a full outer shell and the charge of the metal ion
  - Sodium is in **Group 1,** it will lose 1 electron and form an ion with a **1+ charge**
  - Magnesium is in **Group** 2, it will lose 2 electrons and form an ion with a **2+ charge**
- For non-metals in Group 7 and 6, they will gain 1 and 2 electrons respectively to gain a full outer shell
  - E.g. non-metals in **Group 7** gain 1 electron to form ions with a **1– charge**
  - Non-metals in **Group 6** gain 2 electrons to form ions with a **2– charge**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



All elements are arranged in the order of increasing atomic number from left to right

Examiner Tips and Tricks

The group number is always labelled on the periodic table but the period number isn't.

Don't forget that hydrogen and helium are in Period 1.

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